

LC 1738: Find Kth Largest XOR Coordinate Value

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1. Intuition & Approach

Instead of standard sums, we use **XOR** operations. Unlike other 2D prefix problems that compute prefix sums, we build a **2D Prefix XOR** matrix.

- Build a 2D prefix XOR grid.
- Store every prefix XOR value.
- Find the k^{th} largest value.

2. Example Walkthrough

Given the matrix:

5	2
1	6

The coordinate (1, 1) XOR means computing the XOR of all elements in the subgrid:

$$(5 \wedge 2) \wedge (1 \wedge 6)$$

3. 2D Prefix Logic

For standard 2D prefix sums, we compute the area using:

$$\text{Whole} = \text{Top} + \text{Left} - \text{Overlap}$$

For 2D XOR, the operations transform simply: **+** becomes $\wedge$ and **-** becomes $\wedge$. This works beautifully because $x \wedge x = 0$, meaning XOR naturally removes duplicate inclusions (the overlap).

$$\text{XOR Subgrid} = \text{Top} \wedge \text{Left} \wedge \text{Overlap}$$

4. Finding the Kth Largest

To efficiently get the k^{th} largest XOR coordinate value on the fly, we can maintain a **Min-Heap** (Priority Queue) of size k .

Time Complexity (TC): $O(N \times M \log k)$

Space Complexity (SC): $O(N \times M)$ for the prefix matrix + $O(k)$ for the Min-Heap.

5. C++ Implementation

```
class Solution {
public:
    int kthLargestValue(vector<vector<int>>& matrix, int k) {
        int n = matrix.size();
        int m = matrix[0].size();

        vector<vector<int>> pref(n, vector<int>(m));
        priority_queue<int, vector<int>, greater<int>> mh;

        for(int i = 0; i < n; i++) {
```

```

        for(int j = 0; j < m; j++) {
            pref[i][j] = matrix[i][j];

            if(i > 0) {
                pref[i][j] ^= pref[i-1][j];
            }

            if(j > 0) {
                pref[i][j] ^= pref[i][j-1];
            }

            if(i > 0 && j > 0) {
                pref[i][j] ^= pref[i-1][j-1];
            }

            mh.push(pref[i][j]);

            if(mh.size() > k) {
                mh.pop();
            }
        }

        return mh.top();
    }
};

```